Seat: TBD PETER ANTEATER PANTEATE

CompSci 161 Diagnostic Exam 4, Fall 2025

DO NOT OPEN EXAM UNTIL INSTRUCTED TO DO SO

SILENCE AND STORE ALL ELECTRONICS

This is a diagnostic exam intended to help you evaluate your readiness for the real exam.

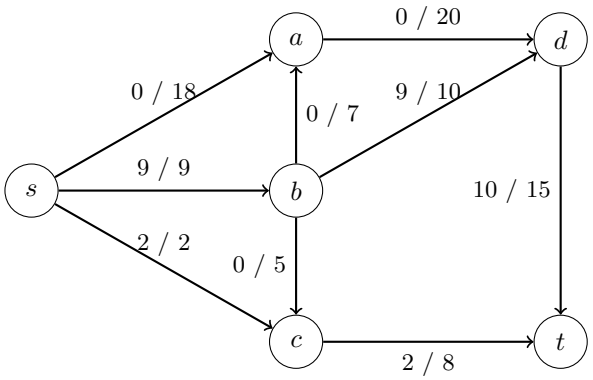
The following rules apply to you, whether you think they do or not. Read and understand them; failure to abide by these rules, or directions given by course staff during the exam, may result in disciplinary action, including but not limited to a failing grade in the class.

- This exam is solely for students enrolled in this lecture. Anyone not enrolled in this lecture may not take an exam.
- You may not open the exam or begin writing until the instructor has explicitly given you permission to do so.
- Keep acceptable identification readily accessible during the test. Proctors may request to see it.
- This exam is closed book, closed notes, and is individual effort. Once course staff begin passing out exams, you may not communicate with anyone other than proctors for any reason, nor may you have electronics, including calculators watches and phones, available to you during the test for any reason. **YOU DO NOT NEED A CALCULATOR!**
- If you leave your seat during the test for any reason, your instructor may collect it and deem you to have turned it in. Do not ask proctors for an exemption to this, they are not authorized to grant such.
- You must take the exam in your assigned seat unless the professor (not a TA) tells you otherwise. The instructor will call to cease writing at 6:45 PM, at which point you must immediately cease writing and close the exam. You may not write any further at that point, including finishing one's current sentence.
- If you believe a question is ambiguous, write at least two reasonable interpretations and indicate clearly which one you will be using. Then answer your question with that assumption. Unless your interpretation makes the problem much more trivial than intended, we will grade your response as if one of us had made that clarification.
- The purpose of the real exam is to evaluate how well *you* understand the material presented in the course. It is an academic integrity violation to do anything that subverts the goals of this assessment including, but not limited to, not doing your own work or submitting that of anyone else.
- We will only grade responses marked in the space provided for each question.

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1. The Ford-Fulkerson algorithm is executed on an input graph G, with source s and sink t. Below is the *flow graph* the algorithm has created after 3 iterations. As in lecture, each edge of the flow graph is labeled "X/Y", where X is the current flow on the edge and Y is the capacity of the edge in G.



Complete the Ford-Fulkerson algorithm and answer the following questions:

(a) What the value of the maximum $s - t$ flow in G? Write **only** a single integer as your response in the box.

(b) What is the cost of the minimum $s - t$ cut in G? Write **only** a single integer as your response in the box.

(c) Which vertices are on the s side of the minimum cut? Circle all that are on the s (source) side on the answer page.

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Question 1:

$1(a)$	$1(b)$

$1(c)$ **Circle** the vertices that are on the **source side**:

s a b c d t

. . . . . .

You may use the space below for scratch paper, but nothing you write below the dotted line will be graded.

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2. Suppose you and your friends (a total of n people) all live in a large house together and are planning to carpool to UCI next quarter. However, you all dislike having to drive, so you want to make sure you have a fair arrangement for who is going to drive the large van you all share. To make matters more complicated, not all of you will go to campus every day, and you want to make sure that whoever drives is someone who is going to school that day.

   Here's one way to define fairness: suppose, for each person p_j, we define Δ_j to be the expected number of days that person j would have driven, if every day's driver is chosen uniformly at random from among the people who are going to school that day.

   Suppose you are given as input a list that, for each person, which day(s) they are going to school next quarter. That is, for each of the d days in next quarter, we know who is going to participate in the carpool (as either a driver or passenger). We say that a **fair driving** schedule is one where each person has at most $\lceil \Delta_j \rceil$ days to drive.

   Use **network flow techniques** to determine a fair driving schedule; that is, for each day, determine who is going to drive that day, in such a way that each person j has at most $\lceil \Delta_j \rceil$ days to drive. For full credit, your running time must be a polynomial function of n and d.

   You don't need to know this for the problem, but under this description, there is always a fair driving schedule. You may treat that statement as given.

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Write your answer for question 2 on this page. It is possible that you won't use the entire page.

Scratch page. Nothing you write on this page will be graded.

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